

Performance Comparison of Single- and Dual-Camera Setups for Image-Based Satellite Orbit Determination

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Abstract

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1 Introduction

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2 Methods

2.1 Single-Camera orbit determination

2.1.1 Gaussian Method

The Gaussian method's approach is to numerically fit observations to a physically feasible orbit. The algorithm yields parameters for full orbit characterization, namely semi major axis a , eccentricity e , argument of periapsis ω and plane normal vector n .

A single observer camera is required, whereas three measurements each consisting of position vector p_i , view vector v_i and corresponding timestamp t_i need to be recorded to determine a solution. [1]

2.1.2 Simplified Gaussian Method

The simplified Gaussian method is a self developed approach to numerically fit observations to a physically feasible circular orbit. Its incentive stems from the condition that satellite orbits low above earth's surface possess very low eccentricities and can mostly be considered circular. The algorithm yields parameters for circle-orbit characterization, namely semi major axis a and plane normal vector n .

Similar to the Gaussian method, a single observer camera is required, whereas only two measurements each consisting of position vector p_i , view vector v_i and corresponding timestamp t_i need to be recorded to determine a solution. Hereby, the solution consists of two scalar parameters ρ_1 and ρ_2 which must satisfy the radius equality condition as well as the arc of a circle condition stated as follows:

$$|r_1| = |r_2| \Leftrightarrow \sqrt{(p_1 + \rho_1 v_1)^2} = \sqrt{(p_2 + \rho_2 v_2)^2} \quad (1)$$

$$\arccos\left(\frac{(p_1 + \rho_1 v_1) \cdot (p_2 + \rho_2 v_2)}{|(p_1 + \rho_1 v_1)|^2}\right) \cdot |(p_1 + \rho_1 v_1)|^{\frac{3}{2}} = (t_2 - t_1)\sqrt{\mu} \quad (2)$$

[2]

2.2 Two camera trajectory determination

2.2.1 Full parametric trajectory

An orbital trajectory can be described as parametric curve lying in its orbital plane as follows:

$$r(\theta) = \frac{a(1 - e^2)}{1 + e \cdot \cos(\theta - \omega)} \cdot \begin{pmatrix} \cos(\theta) \\ \sin(\theta) \end{pmatrix} \quad (3)$$

Given three positions in polar coordinates defined by radii r_i and arguments θ_i , the orbital properties a , e and ω for eccentricities $e \neq 1$ and $e \neq 0$ are determined by:

$$\omega = \alpha_{13} - \arctan(G) \quad (4)$$

$$e = \frac{r_1 - r_2}{C_{12} \cdot \cos(\alpha_{12} - \omega)} \quad (5)$$

$$a = \frac{r_1(1 + e \cdot \cos(\theta_1 - \omega))}{1 - e^2} \quad (6)$$

Hereby, G , C_{12} , α_{12} and α_{13} represent terms consisting of r_1 , r_2 , r_3 , θ_1 , θ_2 and θ_3 .

Two observer cameras are required, utilizing means of stereo vision. Three position vectors of the satellite are obtained by three synchronous measurements which yield values for radii r_i , arguments θ_i and the plane normal vector n . [2]

2.2.2 Circular trajectory

This method once again simplifies the orbit to a circular trajectory. Two observer cameras are required, utilizing means of stereo vision. Two position vectors of the satellite are obtained by two synchronous measurements which yield the semi major axis a and the plane normal vector n . [2]

2.3 Experimental setups

2.3.1 Variable observation spans and observer placement

The goal is to create a measurement setup which can generate data sets usable by all presented single- and dual-camera orbit determination models to evaluate performance under different circumstances. All measurement data is captured from the simulation *Space Engine*, which makes the data synthetic. Space Engine is a software that incorporates a virtual universe including the solar system and nearby, real stars among other things [3]. Identical virtual cameras are used in all observer locations and generate rectilinear images with the following properties:

Image Property	Value
Image frame [pixels]	1000 x 1000
Field of view [°]	17 x 17

Table 1: Image properties of each observer camera

In addition, several cameras are placed in strategic locations, so that orbit determination can be performed subject to variable properties.

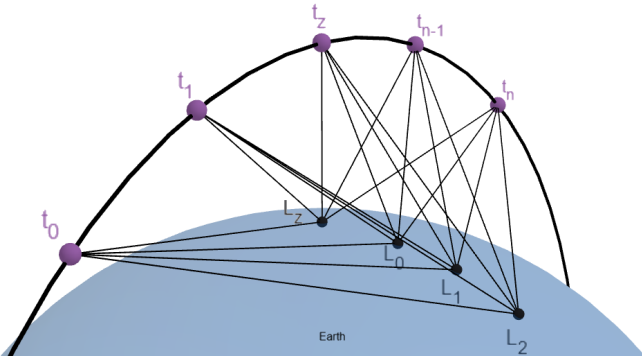


Figure 1: Schematic of strategic camera placement

The schematic figure above shows the figurative placement of cameras as well as the planned synchronous recording of images. Starting at timestamp t_0 and proceeding with a fixed time interval, each camera takes a picture of the satellite at every time stamp until t_n . For each timestamp from t_0 until t_n and for each observer camera, position vector p_i and view vector v_i are recorded

with the Earth Centered Inertial (ECI) coordinate system as reference frame.

As single- and dual-camera orbit determination methods for full determination require only three measurements each and for circular methods only two measurements respectively, we gain some modifiable parameters, which can be altered to observe changes in performance and accuracy of the models.

The first one, which is applicable to single- and dual-camera methods, being the alteration of observation time span Δt which translates to the angular passage $\Delta\theta$ which the satellite passes in its orbit.

Secondly, applicable to dual-camera methods, is the variation in distance between two observer cameras which translates to a variable angle between intersecting lines of sight in the context of stereo vision. Consequently, the observer cameras are placed in the following locations on planet Earth within the simulation software:

Location	Latitude	Longitude	Altitude
L_z	-27.257356°	-64.963269°	313 m
L_0	-27.301281°	-63.939256°	157 m
L_1	-27.339896°	-63.847839°	153 m
L_2	-27.378452°	-63.756358°	155 m
L_3	-27.455384°	-63.573205°	149 m
L_4	-27.608528°	-63.206133°	143 m
L_5	-27.911902°	-62.468913°	127 m
L_6	-28.506740°	-60.982091°	64 m
L_7	-29.646743°	-57.958463°	64 m

Table 2: Observation locations

Hereby, it must be noted that given altitude values may deviate from real ones because these were measured within the simulation software. Cameras in locations from L_0 to L_7 are placed in a straight line, whereas the distance on the ground between location L_i and L_0 doubles for each further location. The distance between L_1 and L_0 equates to 10km, between L_2 and L_0 to 20km and so on up until the distance between L_7 and L_0 equating to 640km. The purpose of location L_z is to observe the satellite passing in the observer's zenith at timestamp t_z . It is not located on line L_0L_7 .

Thirdly, applicable to single-camera methods, is the variation in distance to the orbital plane. Excluding special cases, the distance between observer locations and the orbital plane is constantly changing due to Earth's rotation. However, if the satellite is traversing quick enough from the observer's perspective, an average distance value can be used. Then, a ground truth orbit must be defined, applicable to all presented orbit determination models with the following properties:

Ground Truth Orbit Property	Value
Semi major axis [km]	7067.445
Eccentricity	0
Inclination [°]	52
Ascending Node [°]	0
Reference epoch	J2000
Initial mean anomaly θ_0 [°]	0

Table 3: Ground truth orbit properties

Consequently, the orbit should be circular. It is tilted against the equatorial, whereat the plane normal vector derives from *inclination* and *ascending node* as $n = (0.0, -0.78801075, 0.61566148)$.

Furthermore, it is noted that location L_z was chosen for the predefined timestamp $t_z = 2000.01.01\ 20:03:00$ [UT], so that the satellite passes in the observer's zenith with given ground truth properties.

2.4 Influence of star-based view vector determination

As elaborated in the preliminary text, view vector determination relies three known stars or rather their view vectors v_{S1} , v_{S2} and v_{S3} to form a matrix M for triangulation [2] which is more or less suited for view vector determination through triangulation. One metric to measure suitability is the determinant d as it indicates linear dependency of its row/column vectors formulated as follows:

$$d = \det(M) \quad (7)$$

Another useful metric is the condition number $\kappa(M)$ which indicates how sensitive the triangulation matrix M is to errors formulated as follows:

$$\kappa(M) = \|M\|_2 \cdot \|M^{-1}\|_2 \text{ with } \|M\|_2 = \max_k \sigma_k \quad (8)$$

Hereby, the 2-norm of matrix M is defined by its maximum singular value σ_k [4].

We hypothesize that there might be a correlation between the accuracy of a predicted orbit and the suitability of matrix M used for view vector determination. For testing purposes, we perform orbit determination of the ground truth orbit defined in 3 using distinct sets of stars for view vector calibration. Observations are conducted by one camera placed at L_5 starting at certain predefined time stamps so that numerous background stars are visible to form many distinctive sets of stars for calibration. The determination algorithm of choice is the simplified Gaussian method with a fixed observation span $\Delta t = 7s$.

2.4.1 Influence of variable Semi Major Axis

The goal of this setup to verify whether the self developed simplified Gaussian method holds under differing conditions, specifically with varied semi major axis. Other experiments use solely the ground truth orbit defined in

table 3. This experiment modifies the orbit's semi major axis and initial mean anomaly θ_0 so that the satellite always passes in the zenith of L_z and t_z which makes it easily visible from all observer locations. The semi major axis is chosen in such a way that the range of mean altitudes cover the entire range of the Low Earth Orbit as follows:

Semi major axis	Mean altitude	θ_0
6667.445 km	300 km	198.86°
6867.445 km	500 km	282.36°
7067.445 km	700 km	0.00°
7267.445 km	900 km	72.33°
7467.445 km	1100 km	139.85°
7667.445 km	1300 km	203.00°
7867.445 km	1500 km	262.16°
8067.445 km	1700 km	317.67°
8267.445 km	1900 km	9.85°
8467.445 km	2100 km	58.97°

Table 4: Modified properties of ground truth orbit

Once again, observations are conducted by one camera placed at L_5 with a fixed observation span $\Delta t = 7s$.

2.5 Metrics

2.5.1 Parameter stability

All presented orbit determination methods yield values for semi major axis a and plane normal vector n . Additionally, methods not assuming a circular orbit, yield values for eccentricity e and argument of periapsis ω . The goal is to compare measured (denoted m) with ground truth (denoted g) properties in a generalized form independent of units. For this reason, we define error metrics f_a , f_e , f_ω and f_n for a , e , ω and n as follows::

$$f_a = \frac{|a_g - a_m|}{a_g} \quad (9)$$

$$f_e = |e_g - e_m| \quad (10)$$

$$f_\omega = |\omega_g - \omega_m| \quad (11)$$

$$f_n = \arccos\left(\frac{n_g \cdot n_m}{|n_g| \cdot |n_m|}\right) = \arccos(n_g \cdot n_m) \quad (12)$$

Hereby, specifically for metric 11, it must be noted that this expression is undefined if ground truth or measured orbit has an eccentricity of $e = 0$.

2.5.2 Propagation error

The determined orbit of a satellite differs from the actual ground truth by a margin of error. As the satellite continues orbiting Earth, actual and predicted position continue drifting apart due to this error. Carpenter et al. state that the position error along the trajectory arises from the semi major axis error. This formulates as an

equation for the variation in orbital period δT subject to variation in semi major axis δa as:

$$\delta T = 3\pi \sqrt{\frac{a}{\mu}} \cdot \delta a \quad (13)$$

[5]

Our approach is to formulate the position or rather propagation error in such a way that factors in all orbital parameters. The previous section quantifies the error of individual properties.

3 Results and Discussion

3.1 Parameter error

3.2 Error propagation

3.3 Lower altitude bound for feasible trajectory determination

3.4 Stability of circular models applied to elliptical orbits

4 Conclusion

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